

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1516

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I R.Jaideep (192525441) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R.Jaideep

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong	Reasonable reasoning	Basic reasoning	Weak reasoning

		architecture-level reasoning			
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1. Explain the difference between HDFS and traditional file systems.

Answer: HDFS (Hadoop Distributed File System) is designed to store very large datasets across multiple computers, whereas traditional file systems generally store data on a single computer or storage device. HDFS divides large files into blocks and distributes them across multiple DataNodes. It also maintains multiple copies of the blocks for fault tolerance.

Architecture-level reasoning: In HDFS, the NameNode manages metadata, while DataNodes store the actual data. If one DataNode fails, another DataNode can provide the required data. Therefore, HDFS provides better scalability and fault tolerance for large datasets.

2. Why is replication important in distributed storage systems?

Answer: Replication means maintaining multiple copies of the same data on different machines. It is important because hardware and network failures are common in distributed systems. For example, HDFS commonly uses a replication factor of 3.

Architecture-level reasoning: If one DataNode fails, another DataNode can provide the replica. Rack-aware replication can also protect data from an entire rack failure. Therefore, replication improves data availability, reliability, and fault tolerance.

3. How does MapReduce divide work between map and reduce phases?

Answer: MapReduce divides a large data-processing task into Map and Reduce phases. The Map phase reads input data and generates intermediate key-value pairs. The Shuffle and Sort phase groups the intermediate data according to keys. The Reduce phase processes each group and produces the final result.

For example, in word counting:

Map: <word, 1>

Reduce: <word, total_count>

Architecture-level reasoning: Multiple mapper and reducer tasks can run in parallel on different cluster nodes, allowing very large datasets to be processed efficiently.

4. What role does YARN play in large-scale Hadoop processing?

Answer: YARN stands for Yet Another Resource Negotiator. It is the resource management and job scheduling component of Hadoop. YARN manages CPU and memory resources, application scheduling, resource allocation, task execution, monitoring, and recovery.

Architecture-level reasoning: The ResourceManager manages overall cluster resources, while NodeManagers manage resources and containers on individual nodes. This allows applications such as MapReduce, Spark, and machine learning jobs to share the same Hadoop cluster.

5. Differentiate NAS and SAN in an enterprise data environment.

Answer: NAS (Network Attached Storage) provides file-level storage and is mainly used for file sharing and general-purpose storage. It is relatively easy to configure and manage.

SAN (Storage Area Network) provides block-level storage through a dedicated storage network. It is more complex and expensive but is suitable for high-performance enterprise applications.

Architecture-level reasoning: NAS presents storage as files and directories, while SAN provides storage blocks that servers can access similarly to locally attached disks.

6. What is the purpose of ETL in a big data pipeline?

Answer: ETL stands for Extract, Transform, and Load. Extract collects data from different sources. Transform cleans, validates, formats, and processes the data. Load stores the processed data in a target system such as HDFS, a data warehouse, or a data lake.

Architecture-level reasoning: ETL converts raw data from different sources into a consistent and useful format. This improves data quality and makes the information suitable for analytics and machine learning.

7. How does Apache NiFi help in data integration workflows?

Answer: Apache NiFi is a data-flow and integration tool used to move and process data between different systems. It provides data collection, routing, transformation, error handling, retry mechanisms, monitoring, security, and data provenance.

A typical flow is:

Data Sources → Apache NiFi → Validation/Transformation → HDFS/Data Lake → Analytics

Architecture-level reasoning: NiFi provides controlled and visual data movement, making complex data-ingestion workflows easier to design, monitor, and manage.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer: Major failure points include DataNodes, NameNodes, NodeManagers, ResourceManager, network connections, and storage devices.

Failures can be handled using HDFS replication, NameNode High Availability, ResourceManager High Availability, rack-aware replica placement, continuous monitoring, automatic task rescheduling, backups, and snapshots.

Architecture-level reasoning: Hadoop uses a distributed and fault-tolerant architecture. Data and processing are distributed across multiple nodes, so failure of one component should not bring down the entire system.

9. How would you design a secure data ingestion pipeline for a large organization?

Answer: I would design the pipeline as:

Data Sources → Apache NiFi/ETL → Secure Transfer → Data Validation → HDFS/Data Lake → Processing → Analytics

The main security mechanisms would include TLS/SSL encryption, authentication, role-based access control, encryption of sensitive data at rest, input validation, audit logging, secure credential management, and least-privilege access.

Architecture-level reasoning: Security should be implemented throughout the complete data lifecycle, including ingestion, transfer, storage, processing, and access.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer: I would explain Hadoop using a simple example: “Imagine that a company has one huge warehouse containing millions of files. Instead of keeping everything in one location, Hadoop distributes the data across many storage machines. At the same time, multiple computers can work on different portions of the data.”

The basic architecture is:

Data Sources → HDFS → YARN → MapReduce/Spark → Analytics

HDFS stores large amounts of data, YARN manages computing resources, MapReduce or Spark processes data in parallel, and tools such as Hive help analyze the results.

The main advantage is **scalability**. As data and workload increase, additional machines can be added to the Hadoop cluster. This provides a flexible, reliable, and cost-efficient solution for large-scale data processing.